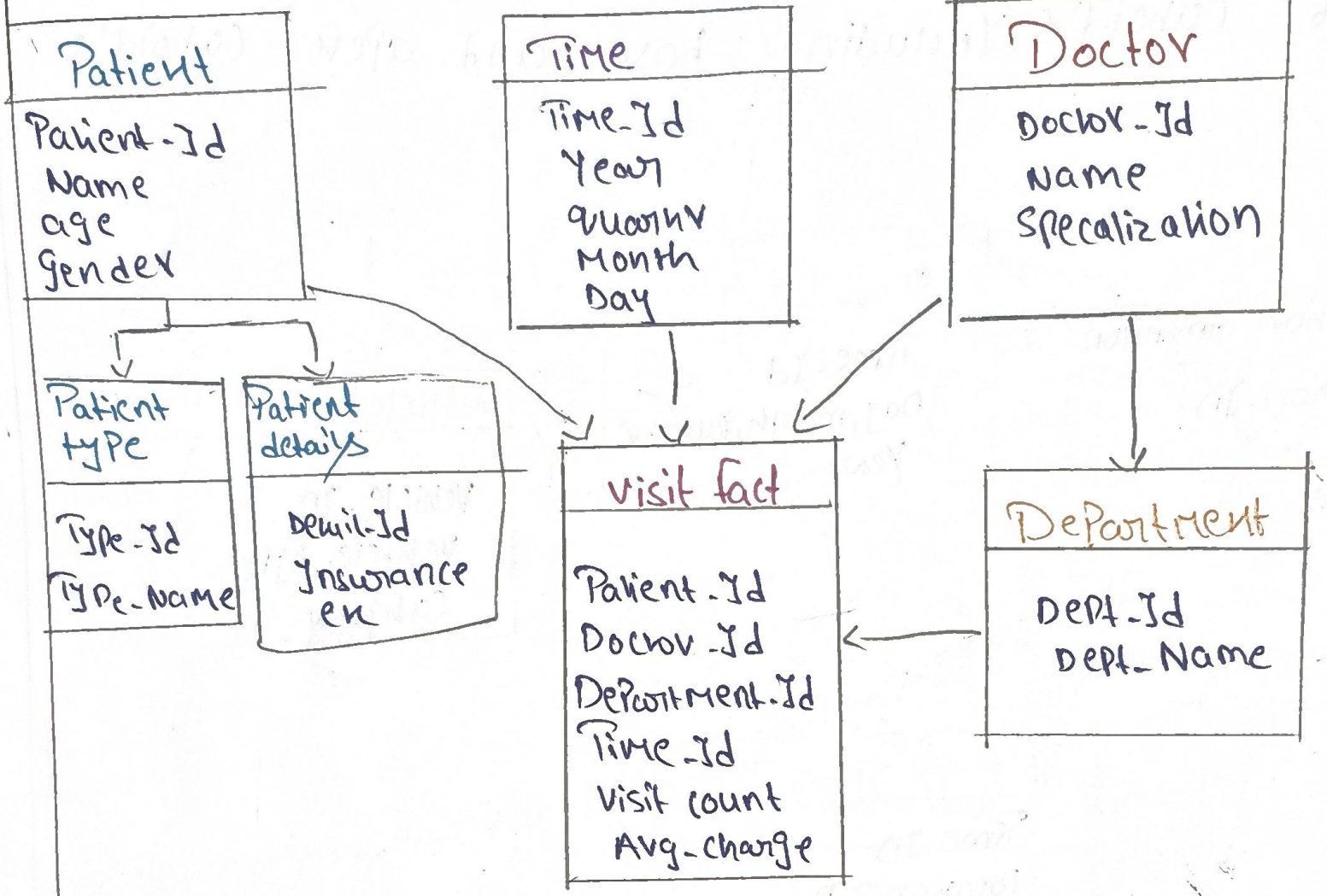
Col-ATI

1A) a) Snowflake Schema



b) OLAP operation's

Starting from base cuboid :-

Patient, Doctor, Department, Time

1. Roll-up Patient → aggregate Patient-level data
2. Roll up Doctor → aggregate doctor-level data
3. Roll up time → move from detailed time to year
4. Keep Department at required level
5. APPLY Avg for each department and year

C. Number of Cuboid's

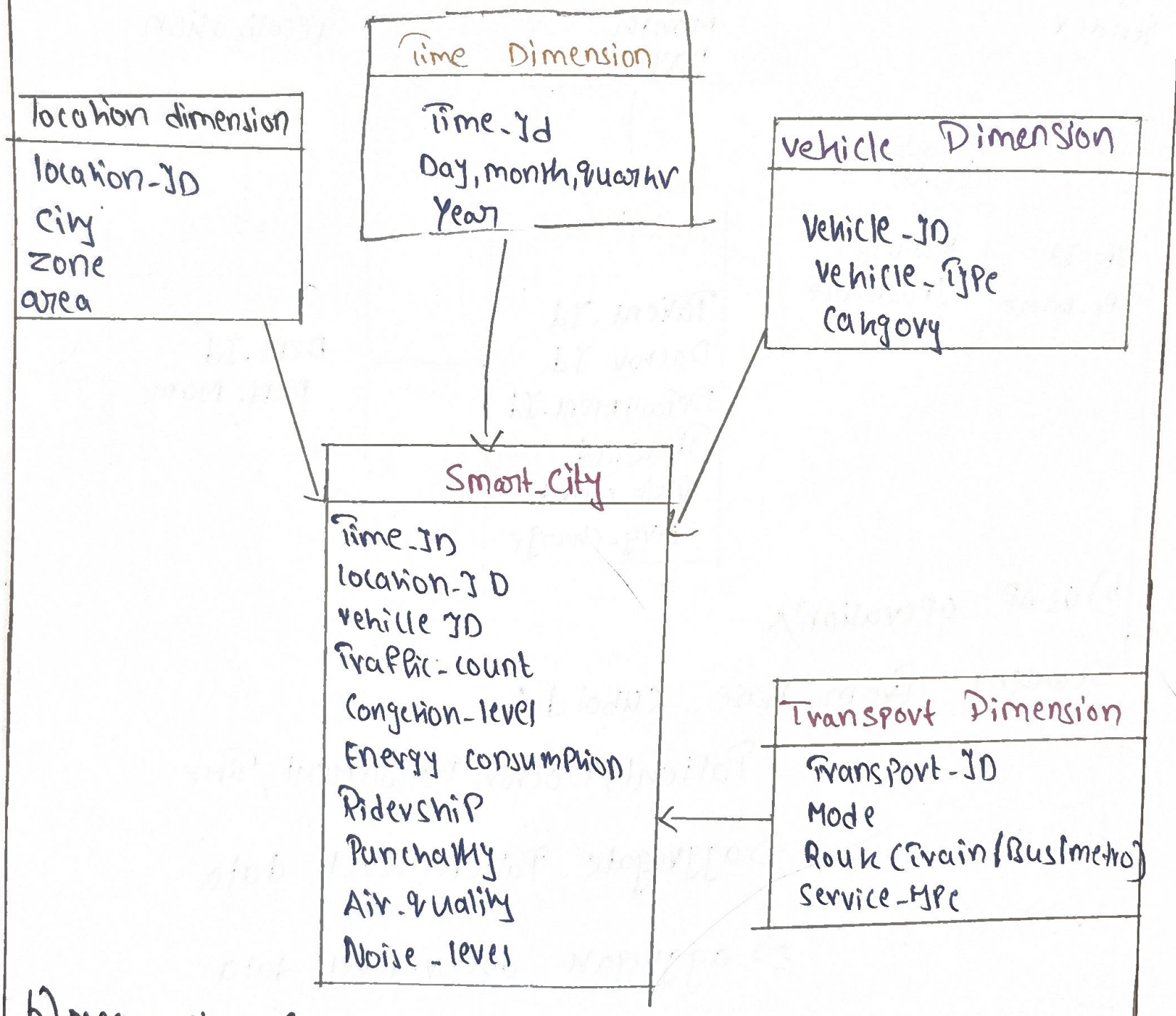
Each dimensions has 4-levels

Number of cuboid's

$4^4 = 256$

256 cuboid's, Including base and apex cuboid's

2A)

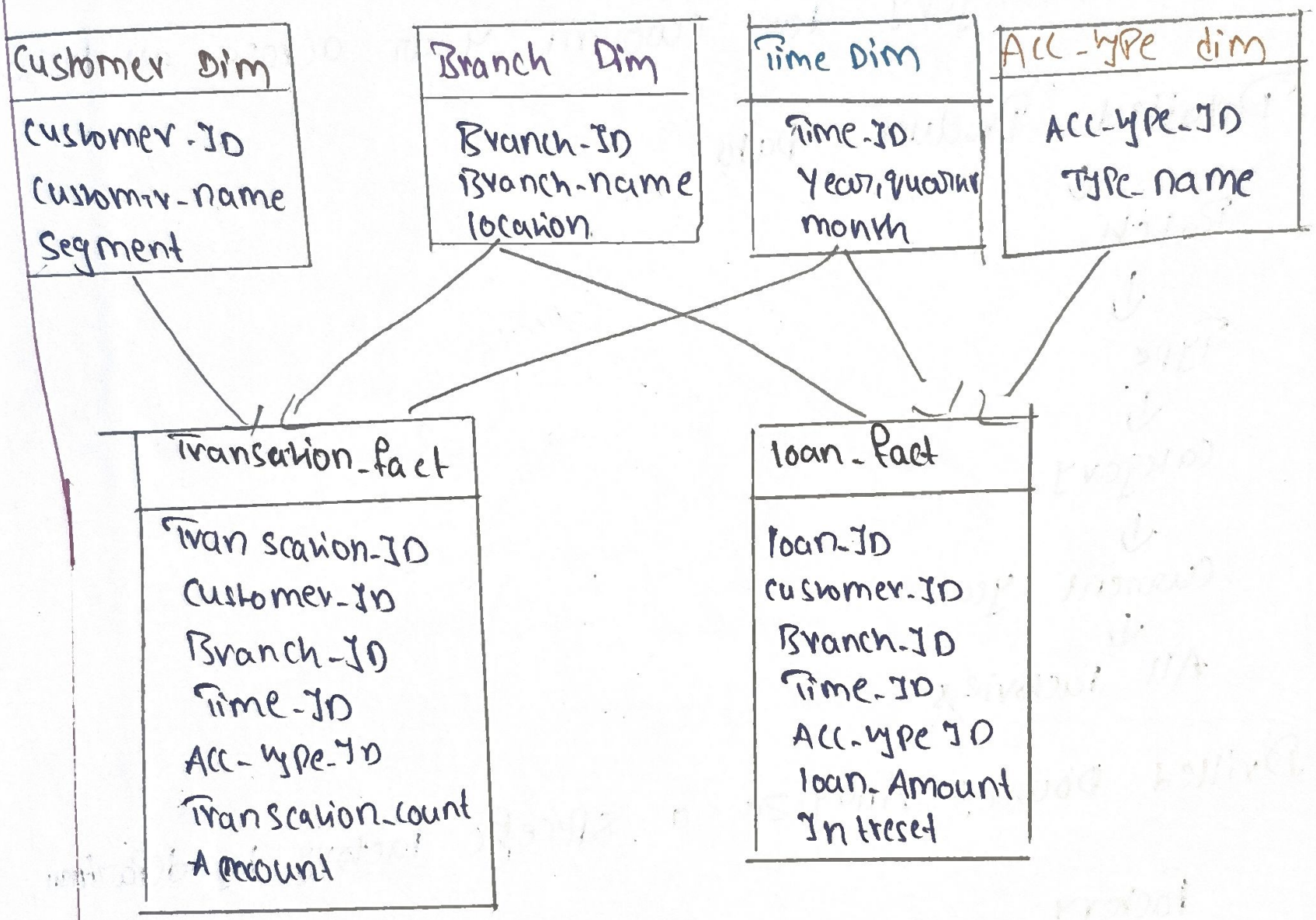


b) Measure's & Dimension's

Measures	Dimension.
* Vehicle count	* Time
* Congestion level	* location
* Energy consumption	* Vehicle
* Air quality	* Transport
* Noise level	

Justification:- Star Schema is simply easy to analyse and works well for multiple related measures and descriptive dimensions

Galaxy Schema



b) OLAP operation

Base Cuboid :- Customer, Time, Branch

Roll up Customer :- aggregate customer level records

Roll up time :- Time to year level

keep branch :- Branch at level

c) Num of cuboid:-

$4^4 = 256$

Total of 256 cut

4A) Roll-up :- Summarize defect rates by Product category for current year across all factories

Detailed Product Data

↓

Batch

↓

Type

↓

category

↓

current year

↓

All factories

Drilled Down :- Anayize a specefic factory by departmen

factory

↓

specefic factory

↓

Department

↓

Product type

↓

Defect Rate

3. Slice :- Production units for suppliers having a valing Ht across All region.

So, Slice selects one condition from supplier dimension while keeping other dimensions available

4. Dice :- Anaylse defect ran and supplier relability for selecnd Product catgories factories location

Measures selecnd:

* Defect Ran.
* Supplier Realiability.

5A) 1. Roll-up :- Summarize invenbory turnover by Product catgory across all stoves for last quarter

SKU
↓
Sub catgory
↓
Product category
↓
last quarter
↓
All stoves

2. Drill down :- Anaylze electronics sales in a

specific region by subcategory and SKU

Electronics.

↓

Specific region

↓

Sub category

↓

SKU

↓

Sales revenue

Drill down Provides Increasingly detailed sales Information

3. Slice :- Sales revenue for male customer in Bronze membership tier across all stores

Condition :-

Gender = Male

Membership tier = Bronze

4. Dice Analyse sales revenue units sold for selected dates Product categories during specific Periods

Ex :-

Cities → Chennai, Bangalore

Product categories → Electronics

Clothing

Time → selected months